



Secure Exit

SIMATS Online Programming Lab

JAVA PROGRAMMING

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QUESTIONS

Q1

Food Delivery Billing System (Multiple Inheritance using Two Interfaces)

10 marks

Q2

Hotel Billing System (Multiple Inheritance using One Class and One Interface)

10 marks

2/2 answered

Q1 Food Delivery Billing System (Multiple Inheritance using Two Interfaces)

10 marks

A food delivery application calculates the final bill for a customer's order by considering the food cost, delivery charges, and coupon discounts.

Create two interfaces:

DeliveryCharge with a method calculateDeliveryCharge().

CouponDiscount with a method calculateDiscount().

Create a class FoodOrder that implements both interfaces.

The delivery charge is calculated as follows:

If the food cost is ₹500 or more, the delivery charge is ₹0.

Otherwise, the delivery charge is ₹50.

The coupon discount is calculated as follows:

If the coupon code is SAVE10, the customer receives a 10% discount on the food cost.

Otherwise, no discount is applied.

The final bill is calculated using the formula:

Final Bill = Food Cost + Delivery Charge – Discount

Display the food cost, delivery charge, discount amount, and final bill.

Input Format

Food Cost

The final bill is calculated using the formula:

Final Bill = Food Cost + Delivery Charge – Discount

Display the food cost, delivery charge, discount amount, and final bill.

Input Format

Food Cost

Coupon Code

Output Format

Food Cost : <food_cost>

Delivery Charge : <delivery_charge>

Discount : <discount>

Final Bill : <final_bill>

Sample Input

600

SAVE10

Sample Output

Food Cost : 600.0

Delivery Charge : 0.0

Discount : 60.0

Final Bill : 540.0

Your Code

Input

```
1 import java.util.Scanner;
```

```
Scanner sc = new Scanner(System.in);
```



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```
1 import java.util.Scanner;
2
3 interface DeliveryCharge{
4     double calculateDeliveryCharge(double foodCost);
5 }
6
7 interface CouponDiscount{
8     double calculateDiscount(double foodCost, String couponCode);
9 }
10
11 class FoodOrder implements DeliveryCharge, CouponDiscount{
12
13     public double calculateDeliveryCharge(double foodCost){
14         if(foodCost >= 500)
15             return 0;
16         else
17             return 50;
18     }
19     public double calculateDiscount(double foodCost, String couponCode){
20         if (couponCode.equals("SAVE10"))
21             return foodCost * 0.10;
22         else
23             return 0;
24     }
25     public static void main(String[] args){
26         Scanner sc = new Scanner(System.in);
27
28         double foodCost = sc.nextDouble();
```

600
SAVE10

Output

Food Cost : 600.0
Delivery Charge : 0.0
Discount : 60.0
Final Bill : 540.0

Visible Test Cases

Test Case 1 ✓
Test Case 2 ✓
Test Case 3 ✓

10.00 / 10 marks



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```
22     else
23         return 0;
24     }
25     public static void main(String[] args){
26         Scanner sc = new Scanner(System.in);
27
28         double foodCost = sc.nextDouble();
29         sc.nextLine();
30         String couponCode = sc.nextLine();
31
32         FoodOrder f = new FoodOrder();
33
34         double deliveryCharge = f.calculateDeliveryCharge(foodCost);
35         double discount = f.calculateDiscount(foodCost, couponCode);
36         double finalBill = foodCost + deliveryCharge - discount;
37
38         System.out.println("Food Cost : " + foodCost);
39         System.out.println("Delivery Charge : " + deliveryCharge);
40         System.out.println("Discount : " + discount);
41         System.out.println("Final Bill : " + finalBill);
42
43         sc.close();
44     }
45 }
```

Submitted



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QUESTIONS

Q1
Food Delivery Billing System
(Multiple Inheritance using Two Interfaces)
10 marks

Q2
Hotel Billing System (Multiple Inheritance using One Class and One Interface)
10 marks

2/2 answered

Q2 Hotel Billing System (Multiple Inheritance using One Class and One Interface)

10 marks

Problem Statement

A hotel wants to generate the final bill for its guests by adding a service charge to the room rent.

Create a class Room that stores the following details:

Room Number
Guest Name
Room Rent

Create an interface ServiceCharge with a method:

```
double calculateServiceCharge();
```

Create a class HotelBill that extends the Room class and implements the ServiceCharge interface.

The service charge is calculated as follows:

If the room rent is ₹5000 or more, the service charge is 12% of the room rent.
Otherwise, the service charge is 8% of the room rent.

The final bill is calculated as:

Final Bill = Room Rent + Service Charge

Display the room details, room rent, service charge, and final bill.

Input Format

Final Bill = Room Rent + Service Charge

Display the room details, room rent, service charge, and final bill.

Input Format

Room Number

Guest Name

Room Rent

Output Format

Room Number : <room_number>

Guest Name : <guest_name>

Room Rent : <room_rent>

Service Charge : <service_charge>

Final Bill : <final_bill>

Sample Input

205

Ananya

6000

Sample Output

Room Number : 205

Guest Name : Ananya

Room Rent : 6000.0

Service Charge : 720.0

Final Bill : 6720.0

Your Code

```
1 import java.util.Scanner;
2
3 interface ServiceCharge {
4     double calculateServiceCharge();
5 }
6 class Room {
7     int roomNumber;
8     String guestName;
9     double roomRent;
10
11     public static void main(String[] args){
12         Scanner sc = new Scanner(System.in);
13
14         HotelBill h = new HotelBill();
15
16         h.roomNumber = sc.nextInt();
17         sc.nextLine();
18         h.guestName = sc.nextLine();
19         h.roomRent = sc.nextDouble();
20
21         double serviceCharge = h.calculateServiceCharge();
22         double finalBill = h.roomRent + serviceCharge;
23
24         System.out.println("Room Number : " + h.roomNumber);
25         System.out.println("Guest Name : " + h.guestName);
26         System.out.println("Room Rent : " + h.roomRent);
27         System.out.println("Service Charge : " + serviceCharge);
28         System.out.println("Final Bill : " + finalBill);
29     }
30 }
31 class HotelBill extends Room implements ServiceCharge{
32
33     public double calculateServiceCharge(){
34         if (roomRent >= 5000)
35             return roomRent * 0.12;
36         else
37             return roomRent * 0.08;
38     }
39 }
```

Input

220
Ana
6000

Output

Room Number : 220
Guest Name : Ana
Room Rent : 6000.0
Service Charge : 720.0
Final Bill : 6720.0

Visible Test Cases

Test Case 1	✓
Test Case 2	✓
Test Case 3	✓

10.00 / 10 marks

```
14 HotelBill h = new HotelBill();
15
16 h.roomNumber = sc.nextInt();
17 sc.nextLine();
18 h.guestName = sc.nextLine();
19 h.roomRent = sc.nextDouble();
20
21 double serviceCharge = h.calculateServiceCharge();
22 double finalBill = h.roomRent + serviceCharge;
23
24 System.out.println("Room Number : " + h.roomNumber);
25 System.out.println("Guest Name : " + h.guestName);
26 System.out.println("Room Rent : " + h.roomRent);
27 System.out.println("Service Charge : " + serviceCharge);
28 System.out.println("Final Bill : " + finalBill);
29 }
30 }
31 class HotelBill extends Room implements ServiceCharge{
32
33     public double calculateServiceCharge(){
34         if (roomRent >= 5000)
35             return roomRent * 0.12;
36         else
37             return roomRent * 0.08;
38     }
39 }
```

Visible Test Cases

Test Case 1	✓
Test Case 2	✓
Test Case 3	✓

10.00 / 10 marks

Submitted